Prediction Markets with Intermittent Contributions

Michael Vitali, Pierre Pinson, *Fellow, IEEE*

***Abstract*—Although both data availability and the demand for accurate forecasts are increasing, collaboration between stakeholders is often constrained by data ownership and competitive interests. In contrast to recent proposals within cooperative game-theoretical frameworks, we place ourselves in a more general framework, based on prediction markets. There, independent agents trade forecasts of uncertain future events in exchange for rewards. We introduce and analyse a prediction market that (i) accounts for the historical performance of the agents, (ii) adapts to time-varying conditions, while (iii) permitting agents to enter and exit the market at will. The proposed design employs robust regression models to learn the optimal forecasts' combination whilst handling missing submissions. Moreover, we introduce a pay-off allocation mechanism that considers both in-sample and out-of-sample performance while satisfying several desirable economic properties. Case-studies using simulated and real-world data allow demonstrating the effectiveness and adaptability of the proposed market design.**

***Index Terms*—Online learning, forecast combination, prediction markets, robust regression, renewable energy.**

I. Introduction

Growing concerns about climate change and energy independence led to a rapid expansion of solar and wind energy production. Forecasting plays a crucial role in decision-making processes, requiring high-quality data and advanced models. The increase in data availability represents an opportunity to refine these techniques and increase forecast quality, and enable companies to increase profits or reduce costs. However, data ownership issues often hinder progress, as companies are reluctant to share information and to collaborate due to competitive interests and security concerns.

A solution to increase collaboration is to give incentives in exchange for these data. This can be done in different ways, depending on whether the companies are willing to share data. If this is the case, different methods have been proposed for renewable power forecasting in the literature exploiting spatio-temporal relations. For solar power forecasting, a model incorporating exogenous data from different sources was developed [1], while a similar approach has been applied to both wind and solar power forecasting [2]. In wind power forecasting, sparse spatio-temporal models were introduced in [3] and later extended in an online learning context [4]. Furthermore, to avoid the loss of privacy companies could face, privacy-preserving solutions using distributed learning methods have been proposed [5], with further advancements addressing online settings [6, 7]. However, all these solutions assume that the agents are willing to collaborate to improve forecasts and that they act rationally and truthfully. This is not always the case in practice. An alternative approach is to view the problem in a more general framework, through prediction markets, in which companies choose to share their individual forecasts and get rewarded for their contribution to the resulting aggregate forecast.

In this context, prediction markets have been increasingly studied in the last decade, gaining popularity across fields [8], with two types of market being proposed: *(i)* contribution-based and *(ii)* "winner takes it all". The latter was proposed by [9], where only the best solution is rewarded. However, this type of market ignores the fact that forecasts other than the best one can still provide additional information. Instead, a contribution-based market rewards all the participants based on the amount of information they brought into the final forecast (as for the example of [10]).

Nevertheless, existing solutions fail to consider some essential aspects of real-world applications such as: *(i)* real-time implementation, *(ii)* historical contributions of the participants, and *(iii)* the ability to accommodate intermittent participation, meaning that participants may join or exit the market at any time. The first two challenges can be seen as an online learning problem, whilst the third one as a missing data problem. For the latter, the literature has proposed robust variants of linear regression [11, 12] and online imputation strategies [13]. In addition, a pay-off allocation mechanism must be designed to determine how rewards are distributed among sellers, reflecting the informational value they contribute to the final forecast. Also in this case, the allocation must reflect historical contributions rather than only instantaneous performance. In the literature, the most widely adopted approach is the Shapley-value-based allocation [14] (see, for example, [7]) and online versions of Shapley value calculation have also been employed for dynamic pay-off distribution [15]. However, these approaches primarily focus on in-sample pay-offs and fail to account for out-of-sample performance, overlooking the actual contribution of sellers to genuine forecast improvements. Finally, any allocation rule must also satisfy several desirable economic properties to ensure fairness and stability.

We introduce and analyse a prediction market[1] that aims at tackling the challenges described in the above. Building upon [10], our market design accommodates different clients and sellers that interact via a market operator. This results in the following contributions:

- The market operator optimally combine input forecasts, while allowing sellers to enter and exit the market at will. This is done through the use of a robust linear regression model that is able to predict in presence of missing data.

Michael Vitali and Pierre Pinson are with the Dyson School of Design Engineering, Imperial College London, London, United Kingdom (e-mail: m.vitali24@imperial.ac.uk; p.pinson@imperial.ac.uk).

[1]Reproducibility package: https://github.com/MichaelVitali/prediction_markets

Additionally, this approach is extended, for the first time, to operate in an online setting.
- A pay-off allocation is proposed accounting for both in-sample and out-of-sample rewards. For the latter, we use a scoring function that assesses the accuracy of each reported forecast and rewards accordingly. Instead, for the in-sample reward, we use a time-varying Shapley value. This is done to reward consistency and informational value provided by the seller over time.

The remainder of the manuscript is organized as follows: Section II introduces the main concepts of the prediction market design. Then, Section III describes its methodological components, from adaptive robust linear regression to pay-off allocation. It is followed by Section IV, showing different test cases and corresponding results. Finally, Section V concludes the paper and offers perspectives for future work.

II. Preliminaries

A. Prediction Markets

Market-based analytics can be broadly categorized into data markets and information markets, depending on whether the exchanged good is raw data or derived information. We focus on prediction markets, a specific subset of information markets. In prediction markets, participants trade forecasts about uncertain future events. These markets combine individual predictions to render a final forecast, communicated to the buyer. Contributors to prediction markets are rewarded in proportion to the value their individual forecasts add to the final forecast. This is done via a mechanism with formal mathematical guarantees concerning desirable economic properties such as budget balance, symmetry, etc.

B. Market Setup

Our market design involves the interaction of multiple clients and sellers, through a central market operator. In its most general form, the market deals with nonparametric probabilistic forecasts, for a continuous variable of interest, based on a set of quantiles for various nominal levels. We define the participant roles as follows:

- client c_i: individual who requires a forecast for a variable of interest (i.e., wind power generation at a set of lead times). The client may or may not already possess their own forecast; in our setup, we focus on clients without an available forecast. We further assume that the client can evaluate the utility U_t derived from a forecast, for instance in terms of profit gains or cost reductions, which is used as a basis for payment for forecast improvement.
- seller s_i: forecaster willing to provide forecasts for the variable of interest, and in the format required by the client. We assume that each seller is contracted and allowed to miss submitting predictions for a maximum proportion of time, e.g. 10% (which may or may not be at random). We denote the set of agents participating in the market as $\mathbf{S} = \{s_1, s_2, \ldots, s_n\}$.
- market operator: central entity responsible for managing the market through a centralized platform. It enables clients to post forecasting tasks for a variable of interest, while allowing sellers to submit their predictions in response to these tasks. Beyond facilitating this exchange, the operator is also responsible for producing the final forecasts, collecting the payment of the clients, and redistributing rewards among participating sellers following a well-defined allocation rule.

For clarity and simplicity, the following sections of this paper will focus on a setting with a single client interacting with multiple sellers.

C. Overview of Market Operation

Fig. 1 provides an overview of the market and a high-level description of the sequential steps for the nominal level τ. At the start of market operations, the absence of historical data presents a challenge for initializing both the ensemble weights and the payoff mechanism. A practical solution is to initialize the weights $\mathbf{w}_t^{(\tau)}$ uniformly across all participating sellers. In parallel, the Shapley values $\boldsymbol{\varphi}_t^{\mathrm{c}(\tau)}$ are initialized to 0. Following this initialization, the process begins with the market opening a session for a specified task. This session remains open for a limited time, during which sellers may submit their forecasts. If the i-th forecast is not submitted, the model finds the missing information from other forecasters (see III-B for details). Once the session closes, the market operator aggregates the submitted forecasts into a combined prediction and delivers it to the client. After the event occurs, the client reports the realization y_{t+1} to the market operator together with the generated utility U_t. At this stage, the market operator performs two parallel operations: *(1)* updating the model and propagating the new weights forward, and *(2)* computing pay-off allocations so that each seller receives their corresponding reward. In addition, the updated in-sample values are incorporated into the next model run.

D. Forecast combination

We define the set of input forecasts as $\hat{\mathbf{X}}_t = \{\hat{\mathbf{X}}_{1,t}, \hat{\mathbf{X}}_{2,t}, \ldots, \hat{\mathbf{X}}_{n,t}\} \in \mathbb{R}^{n\times k\times m}$, from the set of sellers $\mathbf{S}$, with m the number of quantiles (with different nominal levels) and k the lead time. The forecast provided by the i-th seller is $\hat{\mathbf{X}}_{i,t} = \{\hat{\mathbf{x}}_{i,t+1|t}, \ldots, \hat{\mathbf{x}}_{i,t+k|t}\}$. The market operator generates the aggregated forecasts as a weighted average of input forecasts, for each nominal level τ. The weight w_i (for seller s_i) reflects their relative historical performance.

For simplicity, when describing methodological elements, we focus on the case $k = 1$ and a fixed nominal level τ. The set of input forecasts then reduces to $\hat{\mathbf{x}}_t^{(\tau)} = \{\hat{x}_{1,t+1|t}^{(\tau)}, \ldots, \hat{x}_{n,t+1|t}^{(\tau)}\}$. The combined forecast is given as a convex combination of the input forecasts, i.e.,

$$\hat{y}_{t+1|t}^{(\tau)} = \sum_{i=1}^{n} w_i \hat{x}_{i,t+1|t}^{(\tau)}, \quad \sum_{i=1}^{n} w_i = 1, \quad w_i \geq 0 \qquad (1)$$

In our setup, the weights w_i's are non-negative and sum to one, to improve interpretability. In the general case, such constraints are not strictly necessary: weights could be negative and not sum to one. For a comprehensive overview of the state of the art with forecast combination, we refer the reader [16].

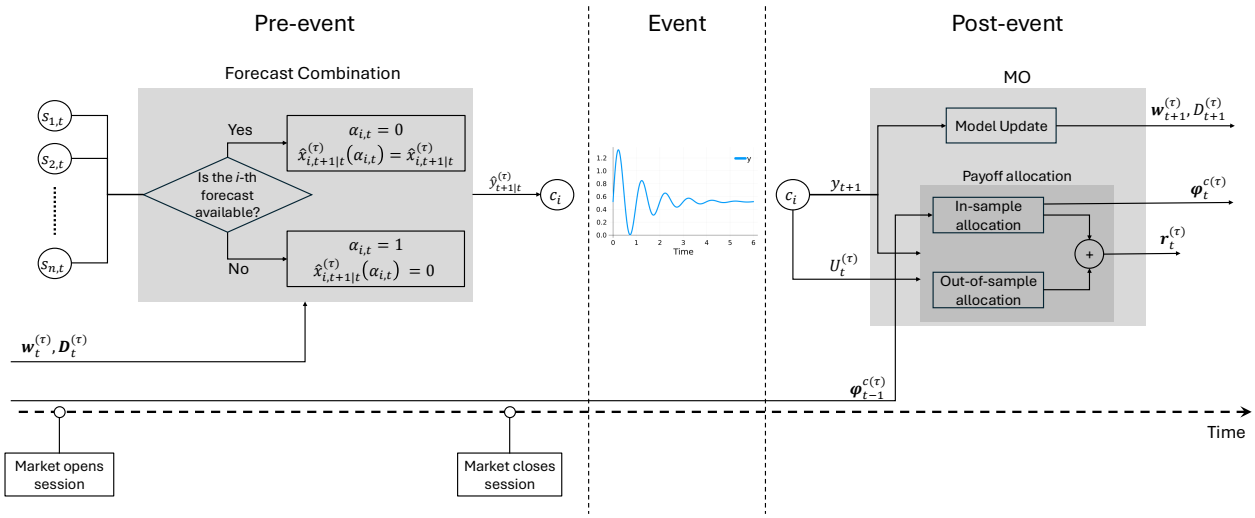

Fig. 1: Market design overview

E. Pay-off allocation

A pay-off function is central to the design of a market mechanism as it distributes the generated utility among the market players (sellers) according to their performances. For this reason, it is critical to design a pay-off function that encourages market participation, whilst reflecting seller's contributions. The pay-off function is characterized by several economic properties that can be mathematically proven, such as budget balance, zero element, etc. Furthermore, the pay-off function can be divided into in-sample and out-of-sample pay-off allocation. The former is used to reward forecast consistency and informational value provided by the sellers in time, whilst the out-of-sample is used to reward good forecasts.

III. Methodology

A. Forecast combination

In this work, the forecast combination is performed using a Linear Regression (LR) model, as introduced in (1). To adapt to potential changes in market dynamics over time, the method is extended to an online learning setting, allowing the model to continuously update and learn the optimal combination of forecasts (weights) in time. For any quantile τ of interest a different LR is learned enabling separate learning dynamics for each level.

The model weights are updated online using online gradient descent (OGD). To derive the update rule, we first define the objective function. In this work, since focusing on nonparametric probabilistic forecasting, we employ the quantile loss (or, pinball loss), defined as

$$\mathcal{L}^{(\tau)}(y,\hat{y}) = \begin{cases} (y-\hat{y})\tau & \text{if } y \geq \hat{y} \\ (\hat{y}-y)(1-\tau) & \text{if } \hat{y} > y \end{cases} \tag{2}$$

where y is the observation, $\hat{y}$ is the aggregated forecast, and τ is the target quantile level. Since the quantile loss is non-differentiable at zero, we utilize the sub-gradient approach studied in [17].

Applying the chain rule to the loss function with respect to an individual weight $w_{i,t}$, we obtain the following sub-gradient:

$$\frac{\partial \mathcal{L}(y_{t+1}, \hat{y}^{(\tau)}_{t+1|t})}{\partial w_{i,t}} = \begin{cases} -\tau \hat{x}^{(\tau)}_{i,t+1|t} & \text{if } y_{t+1} \geq \hat{y}^{(\tau)}_{t+1|t} \\ (1-\tau)\hat{x}^{(\tau)}_{i,t+1|t} & \text{if } \hat{y}^{(\tau)}_{t+1|t} > y_{t+1} \end{cases} \tag{3}$$

Consequently, the weight vector is updated at each step as follows

$$\mathbf{w}^{(\tau)}_{t+1} = \Pi_{\mathcal{H}}\left(\mathbf{w}^{(\tau)}_{t} - \eta \nabla_{\mathbf{w}} \mathcal{L}(y_{t+1}, \hat{y}^{(\tau)}_{t+1|t})\right) \tag{4}$$

where η is the learning rate and $\Pi_{\mathcal{H}}(\cdot)$ denotes the Euclidean projection onto the feasible set $\mathcal{H}$ defined by the constraints in (1). Specifically, $\mathcal{H}$ is the probability simplex

$$\mathcal{H} = \left\{ \mathbf{w} \in \mathbb{R}^n : \sum_{i=1}^{n} w_i = 1, w_i \geq 0 \right\} \tag{5}$$

B. Adaptive Robust Linear Regression

The standard linear combination assumes that sellers submit predictions at every market session. However, this is unrealistic in real-world scenarios. Thus, to allow sellers to participate at will, we adopt a robust variant of the linear regression (LR) model capable of handling missing forecasts. Originally introduced in [11] and applied in [12], this method learns a linear correction matrix $\mathbf{D}^{(\tau)}$ among input forecast, and use this correction to modify the combination weights when some input forecasts are unavailable. In essence, the model compensates for the missing information by extracting additional one from the remaining available forecasts. In this work, we extend this method to the online learning setting.

We model the availability of the forecasts using a binary variable $\alpha_{i,t}$, which takes value 1 if the i-th forecast is unavailable at time t, and 0 otherwise. The forecast vector is redefined such that missing values are zeroed out. Using this, we redefine the forecast vector as

$$\hat{x}^{(\tau)}_{i,t+1|t}(\alpha_{i,t}) = \begin{cases} \hat{x}^{(\tau)}_{i,t+1|t} & \text{if } \alpha_{i,t} = 0 \\ 0 & \text{if } \alpha_{i,t} = 1 \end{cases} \tag{6}$$

The robust forecast combination is then defined as

$$\begin{aligned}\hat{y}_{t+1|t}^{(\tau)} &= [\boldsymbol{\theta}^{(\tau)}(\boldsymbol{\alpha}_t)]^\top \hat{\mathbf{x}}_t^{(\tau)}(\boldsymbol{\alpha}_t) \\ &= (\mathbf{w}^{(\tau)} + \mathbf{D}^{(\tau)}\boldsymbol{\alpha}_t)^\top \hat{\mathbf{x}}_t^{(\tau)}(\boldsymbol{\alpha}_t)\end{aligned} \tag{7}$$

Here, $\mathbf{w}^{(\tau)}$ represents the baseline model when all features are available, while the element $D_{i,j}^{(\tau)}$ represents the linear correction applied to weight $w_i^{(\tau)}$ specifically when the j-th forecast is missing.

We extend this method to the online learning setting by deriving update rules for both $\mathbf{w}$ and $\mathbf{D}$. The gradient for the base weights $\mathbf{w}$ remains as derived in (3). For the correction matrix, we derive the gradient with respect to $D_{i,j}$, which yields

$$\frac{\partial \mathcal{L}(y_{t+1}, \hat{y}_{t+1|t}^{(\tau)})}{\partial D_{i,j;t}} = \begin{cases} -\tau \hat{x}_{i,t+1|t}^{(\tau)} \alpha_{j,t} & \text{if } y_{t+1} \geq \hat{y}_{t+1|t}^{(\tau)} \\ (1-\tau)\hat{x}_{i,t+1|t}^{(\tau)} \alpha_{j,t} & \text{if } \hat{y}_{t+1|t}^{(\tau)} > y_{t+1} \end{cases} \tag{8}$$

Based on these gradients, the parameters are updated iteratively

$$\mathbf{w}_{t+1}^{(\tau)} = \Pi_{\mathcal{H}}\left(\mathbf{w}_t^{(\tau)} - \eta \nabla_{\mathbf{w}} \mathcal{L}(y_{t+1}, \hat{y}_{t+1|t}^{(\tau)})\right) \tag{9}$$

$$\mathbf{D}_{t+1}^{(\tau)} = \mathbf{D}_t^{(\tau)} - \eta \nabla_{\mathbf{D}} \mathcal{L}(y_{t+1}, \hat{y}_{t+1|t}^{(\tau)}) \tag{10}$$

with η learning rate. As in the standard case, a projection step $\Pi_{\mathcal{H}}(\cdot)$ is applied to satisfy the constraints in (1). From (7)-(10), it is evident that the updates depend on the forecast availability vector $\boldsymbol{\alpha}_t$. Specifically, when computing the gradient of the loss function, we observe that the weights corresponding to missing forecasts are not updated, whilst the linear correction matrix is updated only when at least one forecast is missing.

C. Pay-off Allocation

When the client provides the true realization $\mathbf{y}_t$, the total available reward U_t has to be split accordingly to both the in-sample and out-of-sample allocations. The amount of allocation allocated to the two is defined by δ. Let's assume that the total reward is divide equally for each quantile level ($U_t^{(\tau)}$). We have that the reward for the i-th sellers at time t for the quantile level τ is

$$r_{i,t}^{(\tau)} = U_t^{(\tau)}[\delta r_{i,t}^{\text{is}(\tau)} + (1-\delta) r_{i,t}^{\text{oos}(\tau)}] \tag{11}$$

where $r_{i,t}^{\text{is}(\tau)}$ is the in-sample allocation and $r_{i,t}^{\text{oos}(\tau)}$ the out-of-sample one. Finally, the total reward for each seller is defined as $r_{i,t} = \sum_{\tau=1}^{m} r_{i,t}^{(\tau)}$.

1) In-sample allocation: The primary objective of the in-sample allocation is to reward sellers who consistently contribute valuable information to the market. This problem is seen as a cooperative game, where the allocation of the total reward is determined using Shapley values (the definition of which is available at, e.g., [14]). The marginal contribution of seller s_i at time t is defined as

$$\varphi_{i,t}^{\text{s}(\tau)} = \begin{cases} \text{SHAP}_{i,t}^{(\tau)}(y_{t+1}, \hat{\mathbf{x}}_t^{(\tau)}, \boldsymbol{\theta}_t^{(\tau)}) & \text{if } \alpha_{i,t} = 0 \\ 0 & \text{otherwise} \end{cases} \tag{12}$$

If a seller is unavailable at time t, their Shapley value is set to zero. For the remaining sellers, the Shapley values are computed using the corrected weights. In doing so, the method acknowledges that sellers with strong performance helps mitigate the negative impact of missing forecasts on the final aggregated prediction giving them higher value.

However, our goal is to reward sellers not only for their current contributions, but also for the historical informational value they have provided. To achieve this, we employ an online variant of the Shapley value, which is updated over time as

$$\varphi_{i,t}^{\text{c}(\tau)} = \lambda \varphi_{i,t-1}^{\text{c}(\tau)} + (1-\lambda)\varphi_{i,t}^{\text{s}(\tau)} \tag{13}$$

with λ forgetting factor.

Finally, the in-sample reward is calculated as

$$r_{i,t}^{\text{is}(\tau)} = \begin{cases} \frac{\max(0, \varphi_{i,t}^{\text{c}(\tau)})}{\sum_j \max(0, \varphi_{j,t}^{\text{c}(\tau)}) \mathbf{1}_{\{\alpha_{j,t}=0\}}}, & \text{if } \alpha_{i,t} = 0 \\ 0, & \text{otherwise} \end{cases} \tag{14}$$

From the above formula, if the recursive value is negative, it is set to zero. Otherwise, the reward is scaled back considering only the participating sellers so that $\sum_j r_{j,t}^{\text{is}(\tau)} = 1$.

2) Out-of-sample allocation: Differently from the previous allocation, the out-of-sample allocation is used to reward sellers for their instantaneous performance. This is performed using a scoring function

$$sc_{i,t}^{(\tau)} = \begin{cases} 1 - \frac{\mathcal{L}(y_{t+1}, \hat{x}_{i,t+1|t}^{(\tau)})}{\sum_j \mathcal{L}(y_{t+1}, \hat{x}_{j,t+1|t}^{(\tau)}) \mathbf{1}_{\{\alpha_{j,t}=0\}}}, & \text{if } \alpha_{i,t} = 0 \\ 0, & \text{otherwise} \end{cases} \tag{15}$$

where $\mathcal{L}(y_{t+1}, \hat{x}_{i,t+1|t}^{(\tau)})$ is the loss for the i-th forecast. In our framework, the quantile loss is used to evaluate the forecasting accuracy. Similarly to the in-sample allocation, the score for any missing seller is set to zero, while the scores for present sellers are computed exclusively based on the subset of available forecasts.

We have that the out-of-sample reward is defined as

$$r_{i,t}^{\text{oos}(\tau)} = \frac{sc_{i,t}^{(\tau)}}{\sum_j sc_{j,t}^{(\tau)} \mathbf{1}_{\{\alpha_{j,t}=0\}}} \tag{16}$$

3) Properties: In our setting, the pay-off allocation function must satisfy key economic properties to incentivize participation, encourage truthful forecasts, and ensure consistent rewards. We formally define these properties, and sketch the proof of their validity, in the following.

Budget Balance: This property guarantees that the market operator redistributes the exact amount of generated utility among the sellers. For any forecast vector $\hat{\mathbf{x}}_t^{(\tau)}$ and realization y_{t+1}, we verify this by summing the individual rewards:

$$\begin{aligned}\sum_i r_{i,t}^{(\tau)} &= \sum_i U_t^{(\tau)} \left[\delta r_{i,t}^{\text{is}(\tau)} + (1-\delta) r_{i,t}^{\text{oos}(\tau)}\right] \\ &= U_t^{(\tau)} \left[\delta \sum_i r_{i,t}^{\text{is}(\tau)} + (1-\delta) \sum_i r_{i,t}^{\text{oos}(\tau)}\right] \\ &= U_t^{(\tau)}.\end{aligned} \tag{17}$$

Thus, the sum of distributed payments equals the total utility generated.

Symmetry (or, interchangeability): Two sellers who provide identical forecasts must receive identical rewards. Suppose

sellers s_i and s_j provide identical forecasts, then by the symmetry property of the Shapley value, $\varphi_{i,t}^{c(\tau)} = \varphi_{j,t}^{c(\tau)}$. Moreover, identical forecasts yield identical out-of-sample losses, $\mathcal{L}(y_{t+1}, \hat{x}_{i,t+1|t}^{(\tau)}) = \mathcal{L}(y_{t+1}, \hat{x}_{j,t+1|t}^{(\tau)})$, implying equal out-of-sample rewards. Therefore, their final rewards are identical.

Zero-Element (or, dummy agent): Sellers who do not participate should receive no reward. Indeed, if seller i does not submit a forecast at time t, we set $\alpha_{i,t} = 1$. Based on the reward construction in (14) and (15), this explicitly forces the reward to zero, satisfying the property for missing forecasts.

Individual Rationality: This guarantees that no seller is penalized for participating (i.e., pay-offs are non-negative). As observed in (14)-(16), the reward formulation prevents negative values for available sellers, and is exactly zero for unavailable ones. Therefore, $r_{i,t} \geq 0$ holds for all i, t.

Truthfulness (or, incentive compatibility) Sellers maximize their expected reward only by reporting their true forecasts. Following the original work on robust linear regression used for our setting [11], the method can be interpreted as a linear regression model with $d + d^2$ features. Under this interpretation, we can directly apply the truthfulness proof of [18], which shows that altering a feature leads to a strictly higher loss compared to leaving it unaltered. Moreover, it has been established that for linear models, Shapley values preserve truthfulness [18]. Consequently, in our case, the in-sample reward is maximized when the forecast is reported truthfully. Since any alteration also increases the out-of-sample loss, the corresponding reward is lower than that obtained under truthful reporting. Therefore, the overall reward is maximized when the true forecast is provided.

IV. Application and Case-studies

To demonstrate the proposed market and methods, we begin by evaluating the forecasting combination algorithms and pay-off allocation across several examples, starting with two synthetic test cases and concluding with a real-world forecasting scenario. These case studies are, of course, simplified versions of what would be implemented in real-world applications.

A. Synthetic Test Cases

To test the methods proposed in the previous sections, we generated two different environments. First, we implement a time-invariant process that shows the efficacy of the methods proposed and their convergence. Then, a time-varying process is proposed to showcase the ability of the methods to adapt to changes in the dynamics of the environment.

In both scenarios, we consider a single buyer and three sellers. Each forecaster is modeled using a Normal distribution, and the realizations Y_t are generated as a combination of the sellers' distributions. We refer to the standard linear regression model using quantile regression as *QR*, and with *RQR* to the robust implementation. To evaluate performance, we performed a Monte Carlo simulation consisting of 200 independent experiments, with $T = 20000$. For all scenarios, the learning rate of the aggregation method was set to $\eta = 0.01$, the payoff allocation parameter to $\delta = 0.7$, and the forgetting factor to $\lambda = 0.999$.

1) Time-invariant case: The primary goal of this scenario is to verify that the proposed methods works correctly and converge to optimal weights over time. Let $\mu_{i,t}$ and $\sigma_{i,t}$ denote the mean and standard deviation of the i-th seller at time t. The seller's distribution is defined as follows

$$f_{i,t} \sim \mathcal{N}(\mu_{i,t}; \sigma_{i,t}) \tag{18}$$

where $\mu_{i,t} = C_i + \alpha\epsilon_{i,t}$, with C_i constant, $\epsilon_{i,t} \sim \mathcal{N}(0, 1)$. In our setup, we consider three sellers having $C_1 = 0$, $C_2 = 1$ and $C_3 = 2$, $\alpha = 0.5$, and $\sigma_{1,t} = \sigma_{2,t} = \sigma_{3,t} = 1$.

Finally, the realizations are generated as follows

$$Y_t \sim \mathcal{N}(\mu_t; \sigma_t) \tag{19}$$

where $\mu_t = \sum_{i=1}^{n} w_i \mu_{i,t}$, $\sigma_t = \sum_{i=1}^{n} w_i \sigma_{i,t}$, and $\mathbf{w}$ is the vector of weights to be learned. In our case, the weights are set to $\mathbf{w} = [0.1, 0.6, 0.3]$.

Figure 2 illustrates the convergence behavior of both proposed methods (QR top, RQR bottom) with time horizon $k = 1$ and quantile level $\tau = 0.5$. In the RQR case, forecasters are randomly missing with probability 5%. As expected, both algorithms converge over time toward the optimal weights combination.

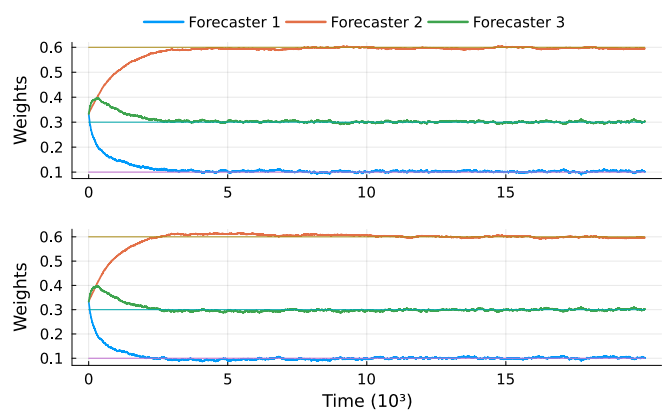

Fig. 2: Convergence of estimated weights for QR (top) and RQR (bottom) with $k = 1$ and $\tau = 0.5$.

Figure 3 shows the pay-off allocation for both methods. We consider three quantile levels, $\mathbf{m} = [0.1, 0.5, 0.9]$, with a total reward of £100 at each time t equally distributed across levels. The final reward of the i-th seller is obtained by summing the rewards for each quantile. As the model weights converge, the reward trajectories stabilize. In the RQR case (bottom plot), the total rewards fluctuate more due to missing forecasts.

2) Time-varying case: In this setting, we simulate a dynamic environment in which the combination of weights evolves over time. The goal is to demonstrate that the proposed methods can detect these changes and adapt accordingly. This property is crucial, as real-world applications are inherently dynamic and subject to temporal variation. To do so, we introduce a periodic coefficient defined as

$$\beta_t = \frac{1}{2}\left(1 + \sin\left(\frac{2\pi t}{T}\right)\right), \quad \beta_t \in [0, 1] \tag{20}$$

Using this coefficient, we define the target weight vector as

$$\mathbf{w}_t^{\text{target}} = (1 - \beta_t)\mathbf{w}^{(1)} + \beta_t \mathbf{w}^{(2)} \tag{21}$$

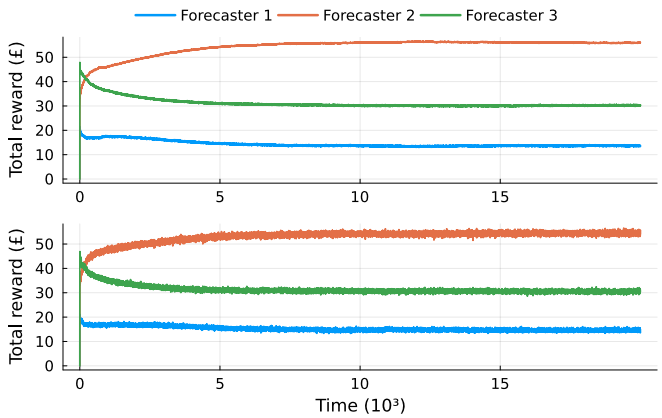

Fig. 3: Pay-off allocation for QR (top) and RQR (bottom) with three quantile levels $m = \{0.1, 0.5, 0.9\}$ and a total step reward of £100.

where $\mathbf{w}^{(1)}$ and $\mathbf{w}^{(2)}$ are two different weight combinations. The actual weights used at time t are then updated recursively

$$\mathbf{w}_t = \lambda \mathbf{w}_{t-1} + (1-\lambda)\mathbf{w}_t^{\text{target}} \tag{22}$$

Finally, the realizations are generated according to (19).

The results shown in Fig. 4 reports the evolution of the estimated weights in a dynamic scenario. Both methods are able to adapt to the evolving environment, with the estimated weights following the underlying periodic pattern. As expected, convergence is not exact, but the overall dynamics are well captured. Also in this case the quantile level considered is $\tau = 0.5$, and for RQR (bottom) the missing rate is 5%.

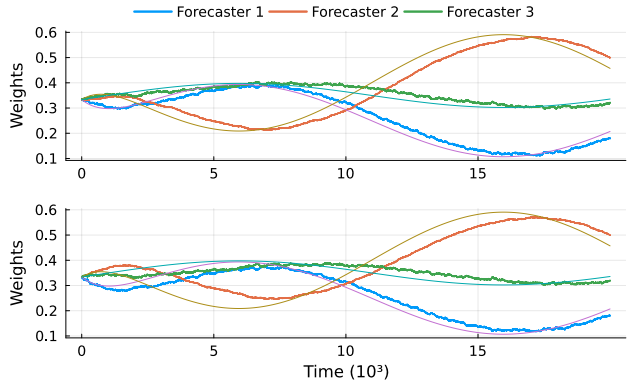

Fig. 4: Convergence of estimated weights for QR (top) and RQR (bottom) with $k = 1$ and $\tau = 0.5$ in a time-varying scenario.

B. Performance with Varying Missingness

In this section, we compare the adaptive QR methods against two benchmark imputation strategies: mean imputation and last-value imputation. The evaluation focuses on the model's tracking performance and how it is affected by different rates of missingness over time. All results are presented for the quantile level $\tau = 0.1$.

The performance of the proposed RQR algorithm was evaluated against two benchmark models, last-impute and mean-impute. Three different sellers were considered with weights w_1, w_2, w_3. The evaluation was based on two key metrics: bias and variance. The results presented in Table I, calculated excluding the first 5000 steps (to consider the results after convergence), show that RQR model has lower variance (~ 1.7) compared to the imputation ones (~ 2.5). This lower variance indicates that RQR's predictions are more stable and exhibit greater consistency across different samples of data. In terms of bias, RQR generally achieved superior or comparable performance. The most significant difference was observed for w_2 and w_3, where RQR's bias was much lower than that of last-impute, whilst having similar results to mean-impute.

TABLE I: Bias and variance for the proposed RQR and the two benchmark models (all values are in units of 10^{-3})

	w_1		w_2		w_3	
	bias	var	bias	var	bias	var
RQR	4.8 ± 3.9	1.7 ± 0.14	-5 ± 6.5	1.7 ± 0.2	0.3 ± 4	1.7 ± 0.16
Last Impute	5 ± 4.6	2.5 ± 0.2	-35 ± 7	2.5 ± 0.3	-18 ± 6	2.5 ± 0.2
Mean Impute	7 ± 5	2.5 ± 0.2	-8 ± 9.4	2.5 ± 0.4	-0.3 ± 0.6	2.5 ± 0.2

Fig. 5 illustrates the bias of the RQR method under varying levels of missingness. The missingness rate ranges from 5% to 90%, with the constraint that at least one seller is always present. The results indicate that, at low missingness rates, the model exhibits an average bias close to zero. As the missingness rate increases, the bias also rises, which is expected since the model becomes less capable of learning the optimal combination and compensating through the linear correction. In terms of variance, however, the differences are minimal, indicating that the model maintains consistent performance even under high missingness conditions.

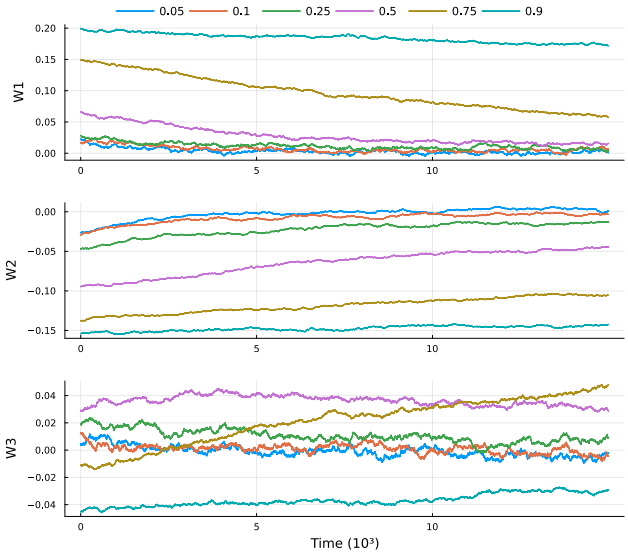

Fig. 5: Bias with varying missingness rate ranging from 5% to 90%.

C. Real-world Forecasting Problem

We demonstrate the proposed market framework through a wind energy forecasting case study. The analysis focuses

on day-ahead probabilistic power forecasting for an offshore wind farm located in Belgium. The power production data are obtained from the open data portal of the Belgian Electricity System Operator, Elia Group [19]. The wind farm has a nominal installed capacity of 2,262.1 MW, and the dataset covers the period from 2024-01-01 to 2025-12-31 at a 15 min resolution. As sellers instead, we include forecasts from three NWP models, namely (i) the High-Resolution Forecast (HRES) from the European Centre for Medium-Range Weather Forecasts (ECMWF) [20], (ii) the Global Forecast System (GFS) from the National Oceanic and Atmospheric Administration (NOAA) [21], and (iii) the ICON-EU from Deutscher Wetterdienst (DWD) [22]. For each NWP dataset, we consider a grid surrounding the wind farm location. These spatial features are subsequently transformed using Principal Component Analysis (PCA) in order to obtain a reduced set of uncorrelated predictors. The resulting pre-processed features are then used to train three different probabilistic forecasting models for each NWP source. Specifically, we employ (i) Extreme Gradient Boosting (XGBoost), (ii) Quantile Regression Forests (QRF), and (iii) a Multi-Layer Perceptron (MLP). Since three models are fitted for each of the three NWP datasets, this results in a total of nine probabilistic forecasts that are subsequently considered as input to the market. It is worth emphasizing that the primary objective of this study is not to develop the most accurate individual forecasting models, but rather to demonstrate the effectiveness of the proposed market framework.

Moreover, the individual forecasting models were trained exclusively on data from 2024. Then, the cross-validation and testing of the online prediction market were performed on data from 2025. To improve convergence and reduce gradient variance when adapting market weights across multiple lead times, we implemented a mini-batch strategy for the online gradient updates, averaging the loss across each batch. Specifically, to determine the best learning rate and batch size for the online prediction market, cross-validation was performed on the first two months of 2025 (January and February). We conducted a grid search over the learning rates $\eta \in \{0.2, 0.1, 0.05, 0.01, 0.005\}$ and batch percentages $b \in \{0.05, 0.1, 0.2, 0.5, 1.0\}$. The optimal hyperparameters for the market mechanism were identified as 0.1 for both the learning rate and the batch size percentage. Final testing was then performed on the remaining ten months of 2025, where the initial two-month cross-validation timeframe was utilized as a burn-in period to initialize the online mechanism.

Finally, to assess the robustness of the results, we conduct a Monte Carlo simulation consisting of 100 independent experiments. The reported performance metrics correspond to the average results over these runs.

Table II shows the quantile loss achieved by all considered methods across the different nominal levels. Overall, models trained using weather forecasts from ECMWF exhibit superior predictive performance compared the others. Of the combination method, QR delivers the best overall performance and serves as an oracle benchmark, as it assumes full information availability. This result confirms that combining forecasts based on heterogeneous information sources leads to improved probabilistic accuracy. The main objective of this work, however, is to design a method that remains effective in the presence of missing data. In this respect, the proposed RQR approach demonstrates strong robustness. Even with 5% missingness, RQR achieves quantile loss values that remain very close to the oracle QR benchmark, while still improving upon the best individual weather-based model. This shows that the method is capable of efficiently exploiting the available information and mitigating the negative impact of partial data unavailability. Finally, as expected, the quantile loss increases monotonically with the missingness rate.

TABLE II: Quantile loss for different nominal levels and methods.

	Model	Quantile		
		0.1	0.5	0.9
ECMWF	QRF	37.76	86.68	45.78
	MLP	37.18	85.45	41.05
	XGB	36.19	82.38	43.92
NOAA	QRF	41.11	100.42	52.24
	MLP	40.13	96.98	49.14
	XGB	38.66	95.96	50.79
DWD	QRF	40.14	96.40	49.59
	MLP	38.13	90.36	44.48
	XGB	38.43	91.55	48.12
Combination	QR	**34.01**	**76.84**	**36.72**
	RQR - 5%	34.79	81.55	39.07
	RQR - 10%	35.59	84.81	40.34
	RQR - 20%	37.43	91.44	45.25

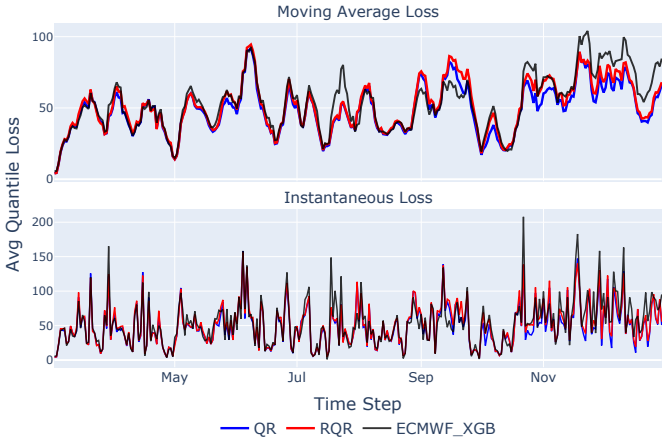

Fig. 6: Moving average and instantaneous loss of the best-performing expert as well as the online combination methods.

Fig. 6 illustrates the instantaneous loss (bottom panel) and seven-day moving average loss (top panel) for the most accurate individual expert (ECMWF-XGB) alongside the combination methods. While the instantaneous loss highlights the high daily volatility of the predictions, it shows that the oracle (QR) and RQR (with 5% missingness) generally mitigate the most severe error spikes compared to the single expert. The advantages of these adaptive combination techniques become particularly evident when examining the moving average loss.

Here, RQR demonstrates robust predictive capability, consistently maintaining a lower moving average loss than the single best forecaster. It becomes clear that adaptive techniques prove highly beneficial in instances where one or more experts underperform; this is notably visible after November, where the moving average loss of the single expert spikes significantly above the combination methods.

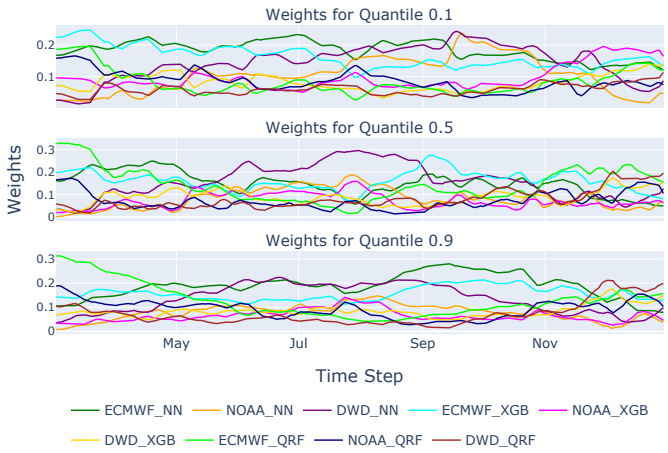

Fig. 7: The evolution of weights for each quantile level over time for the RQR model.

Figure 7 displays the seven-day moving average of the combination weights for each quantile level over the test period. Overall, the largest and most stable weights are assigned to ECMWF-MLP and ECMWF-XGB across all nominal levels, which is consistent with the results in Table II. Interestingly, the DWD-MLP expert exhibits a distinct time-varying behavior. For the lower quantile level (0.1), its weight increases noticeably during the second half of the test period. Moreover, between June and September, DWD-MLP attains the highest weight for both the median (0.5) and upper (0.9) quantiles. This occurs despite its overall performance being less competitive when evaluated across the entire test set.

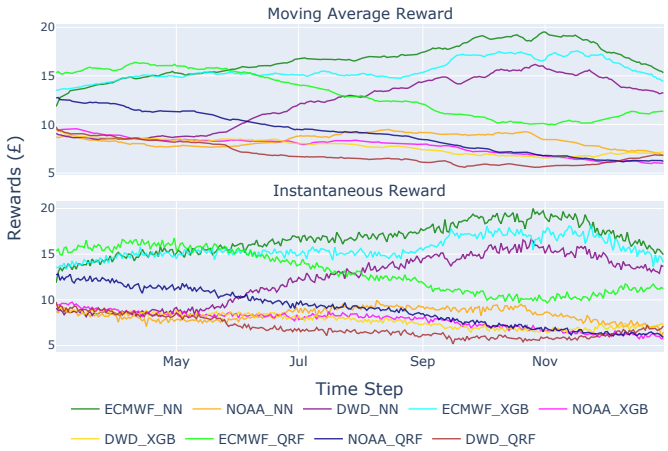

Fig. 8: Moving average and instantaneous rewards for all market participants considering the RQR method.

Fig. 8 illustrates the total reward distribution across all forecasters. Because the reward calculation prioritizes the Shapley value over instantaneous error, the dynamics explicitly emphasize models that yield the most distinct predictive value. Notably, the MLP and XGB models trained on ECMWF data accumulate the highest sustained rewards over time. This demonstrates their ability to consistently deliver substantial unique information while maintaining strong temporal accuracy (see Table I). Additionally, the reward for the DWD-MLP model exhibits a clear upward trend that mirrors the shifts observed in the weight distributions. This growth is likely driven by a concurrent improvement in both the model's accuracy and the unique informational value it contributes to the aggregated forecast.

Finally, Table III details the monthly reward distribution among the nine market participants, supporting the trends observed in the moving average plots. Overall, the ECMWF-MLP and ECMWF-XGB models consistently achieve the highest monthly rewards. The temporal dynamics also reveal distinct performance shifts over the year: notably, DWD-MLP demonstrates a strong upward trend, while ECMWF-QRF exhibits a downward trend.

V. Conclusions

The growing availability of data offers unprecedented opportunities to advance forecasting models and improve the integration of renewable energy generation. Yet, competitive interests and concerns over data privacy often prevent stakeholders from sharing information. There, we proposed a new prediction market platform that enables communication among stakeholders while incentivizing participation through rewards. Our market design considers many real-world application challenges and demonstrates how those can be solved. Several challenges remain open. E.g., the market can be extended to a fully dynamic setting, where the market allows a dynamic set of participants over time. And, alternative pay-off allocation mechanisms beyond Shapley value should be explored to reduce computational complexity and enhance scalability.

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TABLE III: Monthly rewards for different methods (March - December 2025).

	Model	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
ECMWF	QRF	467.70	485.00	485.86	439.90	413.26	383.76	332.02	316.13	308.12	335.02
	MLP	415.65	451.01	478.22	487.03	517.99	530.06	548.49	589.32	558.59	489.87
	XGB	422.05	448.96	471.86	454.84	470.98	466.49	499.70	531.54	516.31	461.82
NOAA	QRF	368.29	343.47	338.99	296.48	288.74	273.87	235.84	219.48	196.40	187.65
	MLP	255.22	235.80	240.33	248.33	277.19	287.12	269.54	278.29	240.93	215.59
	XGB	276.59	255.17	259.37	247.07	255.78	249.98	226.54	215.95	191.80	183.76
DWD	QRF	268.66	251.36	245.26	205.96	204.48	197.92	178.36	179.07	176.04	197.97
	MLP	258.68	259.30	283.65	335.58	391.14	425.74	439.86	487.21	456.57	409.87
	XGB	263.21	254.76	260.77	249.57	247.54	236.76	211.56	209.04	202.87	212.30

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